

AIVOA.AI

AI Integration & End-to-End Workflow

AI, RAG, LangGraph orchestration and human review workflow

Document 5 of 5

Professional presentation edition • Content preserved from the supplied source documents

5. AI Workflow A — Deviation Intake

INPUT PDF / image / pasted narrative + optional batch context 1. File validation 2. Parse text 3. OCR if needed 4. Structured extraction 5. Validate JSON 6. Resolve batch/process context 7. Build retrieval query 8. Retrieve relevant evidence 9. Run deterministic checks 10. Ask LLM for contextual assessment 11. Return review package

6. Structured Extraction Prompt Design

The prompt should force the model to extract only what is supported by the supplied document/context. Missing fields should be null rather than guessed.

SYSTEM You extract structured quality-event data from supplied manufacturing records. Use only the supplied content. Do not invent batch numbers, measurements, dates, causes, or actions. If a field is not supported, return null. Return JSON matching the provided schema.
USER DOCUMENT: {document_text} KNOWN CONTEXT: {batch_context} Return: { "title": ..., "batch_number": ..., "process_step": ..., "parameter": ..., "expected_range": ..., "actual_value": ..., "duration_minutes": ..., "description": ..., "immediate_action": ... }

Pydantic validation should happen immediately after the model response. A syntactically valid JSON object is not necessarily semantically valid, so field-level checks should follow.

7. AI Workflow B — RAG Evidence Retrieval

Document ingestion ↓ knowledge_documents ↓ chunking ↓ knowledge_chunks ↓ embedding ↓ pgvector Runtime ↓ deviation / investigation query ↓ query embedding ↓ top-k similarity search ↓ metadata filtering ↓ evidence package ↓ LLM context

7.1 Retrieval metadata filters

Company/tenant where applicable.

Document type, such as SOP, procedure, batch record or quality procedure.

Document status, preferring approved/effective sources.

Document version/effective date.

Manufacturing site where relevant.

Optional product/process metadata.

7.2 Evidence object

```
{ "document_id": "uuid", "document_title": "SOP-014 v3.2", "chunk_id": "uuid", "source_type": "SOP", "snippet": "Approved temperature range is 76-80 °C...", "relevance": 0.91, "effective": true }
```

The UI should show source title/version and a short snippet. The model should receive the same identifiers so its recommendation can be traced back to evidence.

8. AI Workflow C — Severity / Impact Recommendation

Deterministic inputs ■■■ expected vs actual ■■■ duration ■■■ product/batch status ■■■ predefined risk rules ↓ Risk context + Retrieved evidence ↓ LLM contextual assessment ↓ Structured recommendation ↓ Human review

8.1 Recommendation schema

```
{ "recommended_severity": "major", "recommended_impact": "potential_product_quality_impact", "reason": "Actual temperature exceeded the approved range...", "evidence_ids": ["chunk-1", "chunk-8"], "confidence": 0.87, "requires_human_review": true }
```

Do not expose confidence as if it were a calibrated probability unless the model/evaluation process actually supports that interpretation. In the UI, "AI confidence" is best treated as a model-generated confidence signal, not a regulatory measurement.

9. AI Workflow D — Investigation Assistant

Deviation context + Current investigation tasks + Retrieved SOPs / batch records / prior events ↓ AI Investigation Planner ↓ Suggested: • evidence to review • questions to answer • records to compare • possible contributing factors ↓ Human investigator selects tasks ↓ Tasks become persisted investigation records

9.1 Example output

Suggested investigation checks 1. Review equipment maintenance history for the cooling valve. Evidence: equipment log / maintenance SOP 2. Compare temperature excursion against approved process limits. Evidence: SOP-014 v3.2 3. Review operator response and intervention time. Evidence: batch manufacturing record 4. Assess whether the excursion could affect product quality. Evidence: product/process specification

AI should propose investigation scope, not create completed evidence. A suggested task is still a task, not a fact.

10. AI Workflow E — Root Cause / 5 Whys

Investigation evidence ↓ Evidence summarizer ↓ Candidate cause generator ↓ 5 Whys draft ↓ Evidence support check ↓ Human investigator review ↓ Confirmed root cause

10.1 Prompt structure

SYSTEM You are assisting a quality investigator. Generate candidate root causes only from the supplied investigation evidence. Separate observed facts, inferences, and hypotheses. Do not present a hypothesis as a confirmed root cause. USER PROBLEM: {problem_statement} EVIDENCE: {evidence} OUTPUT: { "candidate_root_causes": [...], "five_whys_draft": [...], "supporting_evidence_ids": [...], "uncertainties": [...] }

10.2 Human review UI contract

Show AI draft and evidence together.

Allow the investigator to edit every Why step.

Require a final root-cause text field.

Record Accept/Edit/Reject.

Record which fields were changed.

Persist the evidence IDs supporting the final result where applicable.

11. AI Workflow F — CAPA Suggestions

Confirmed root cause + Contributing factors + Investigation evidence ↓ CAPA suggestion node ↓ Corrective action candidates Preventive action candidates ↓ Human selection/edit ↓ Persist CAPA actions

11.1 Example

Root cause: Inadequate preventive-maintenance control for cooling-valve actuator. AI candidate actions: Corrective: - Replace faulty actuator. - Verify actuator function before next batch. Preventive: - Update PM checklist. - Add actuator inspection frequency. - Add maintenance record review step. Human reviewer: ✓ Replace actuator ✓ Update PM checklist ✗ Change due date ✗ Ignore inspection-frequency suggestion

The model should not invent due dates or assign owners without supplied business context. Those fields belong to the user/workflow.

12. AI Workflow G — Effectiveness Review

CAPA effectiveness criteria + Observed monitoring data + Linked batches / quality events ↓ AI trend summary ↓ Possible recurrence / no recurrence ↓ Human effectiveness decision

For the demo, the effectiveness plan can be "Monitor next 5 batches". AI can summarize whether the monitored batches show recurrence, but the quality reviewer records the final Effective/Ineffective decision.

13. AI Workflow H — Closure Summary

Deviation + Investigation conclusion + Root cause + CAPA actions + Effectiveness result ↓ AI closure-summary draft ↓ Human review ↓ Authorized closure ↓ Audit event

The closure summary is a convenience feature. Closure authority remains with the authorized quality user.

14. Recommended LangGraph Node Map

15. Pseudocode for the Core Graph

```
graph = StateGraph(QMSAState) graph.add_node("validate_input", validate_input) graph.add_node("parse_document", parse_document)
graph.add_node("ocr_fallback", ocr_fallback) graph.add_node("extract_fields", extract_fields) graph.add_node("validate_extraction",
validate_extraction) graph.add_node("build_query", build_query) graph.add_node("retrieve_evidence", retrieve_evidence)
graph.add_node("deterministic_assessment", deterministic_assessment) graph.add_node("llm_assessment", llm_assessment)
graph.add_node("create_review_package", create_review_package) graph.add_edge(START, "validate_input")
graph.add_edge("validate_input", "parse_document") graph.add_conditional_edges( "parse_document", lambda s: "ocr_fallback" if
s["ocr_used"] else "extract_fields" ) graph.add_edge("ocr_fallback", "extract_fields") graph.add_edge("extract_fields", "validate_extraction")
graph.add_edge("validate_extraction", "build_query") graph.add_edge("build_query", "retrieve_evidence")
graph.add_edge("retrieve_evidence", "deterministic_assessment") graph.add_edge("deterministic_assessment", "llm_assessment")
graph.add_edge("llm_assessment", "create_review_package") graph.add_edge("create_review_package", END)
```

16. Prompt Versioning

Prompts are part of the application behavior and should be versioned. Store a prompt_version with every AI interaction.

17. Grounding Rules

Facts from the user's uploaded quality record take precedence over generic model knowledge.

Controlled SOP/procedure evidence should be retrieved from the QMS knowledge base when available.

The model must distinguish evidence-backed facts from hypotheses or suggestions.

If no supporting evidence exists, say that evidence is unavailable rather than manufacturing a citation.

Do not use web search as the source of internal SOP requirements in the demo. Internal QMS documents are the relevant evidence source.

Prior deviations can be used as contextual examples only if their content is authorized and relevant.

18. AI Guardrails

18.1 Prompt injection handling

A scanned batch record or uploaded document may contain text that looks like an instruction to the AI. Treat all document text as data. The system prompt must explicitly state that instructions found inside source documents are not system instructions. This matters because humans have apparently decided that PDFs should now contain adversarial prompt payloads. Civilization continues.

19. Exact Request Flow: Frontend → Backend → AI → Database

1. User opens DEV / Log Deviation. 2. User uploads temperature_log.pdf. 3. React sends multipart POST /deviations/process. 4. FastAPI validates token and tenant membership. 5. Service stores/refs upload. 6. LangGraph validates input. 7. Parser extracts text. 8. OCR runs if required. 9. Extraction node returns structured event. 10. Pydantic validates extraction. 11. Query builder creates evidence query. 12. pgvector retrieves SOP/batch evidence. 13. Deterministic rules compare expected vs actual. 14. Groq produces contextual recommendation. 15. Pydantic validates recommendation. 16. Backend returns review package. 17. UI shows: extracted fields severity recommendation evidence Accept/Edit/Reject 18. User accepts/edits. 19. React POST /deviations with reviewed values. 20. Backend persists final deviation. 21. Backend persists AI interaction + evidence links. 22. Backend appends audit event. 23. UI shows DEV-2026-018. 24. User clicks Start Investigation. 25. Backend creates INV-2026-012. 26. Investigation AI suggests evidence/tasks. 27. Human investigator completes investigation. 28. AI drafts 5 Whys. 29. Human confirms root cause. 30. CAPA created. 31. AI suggests corrective/preventive actions. 32. Human approves actions. 33. Effectiveness monitoring recorded. 34. AI summarizes monitoring. 35. Human marks effective/ineffective. 36. Authorized user closes. 37. Batch Release screen reflects final quality status.

20. AI API Contract

POST /api/v1/ai/deviation/process Request: multipart/form-data - file - batch_id (optional) - context (optional) Response: { "request_id": "...", "status": "review_required", "extraction": {...}, "assessment": {...}, "evidence": [...], "recommendation_id": "...", "next_action": "human_review" }

20.1 Human review endpoint

POST /api/v1/ai/recommendations/{recommendation_id}/review { "decision": "edited", "edited_fields": { "severity": "major" }, "review_note": "Confirmed after QA review." } Response: { "recommendation_id": "...", "decision": "edited", "record_ready": true }

21. What Gets Stored After Every AI Run

22. Failure / Fallback Workflow

LLM unavailable ↓ Return: extraction = available deterministic_checks = available AI recommendation = unavailable requires_human_review = true RAG unavailable ↓ Return: recommendation may be produced only if safe, but evidence-backed claim = unavailable Malformed LLM output ↓ Pydantic validation fails ↓ One constrained repair attempt ↓ If still invalid → controlled failure ↓ No false success / no invented record

The user should see a clear status such as "AI recommendation unavailable. Manual review required." The system should never quietly replace a failed AI run with a made-up answer.

23. AI Evaluation Checklist for the Demo

24. Recommended 5–10 Minute AI Demo Story

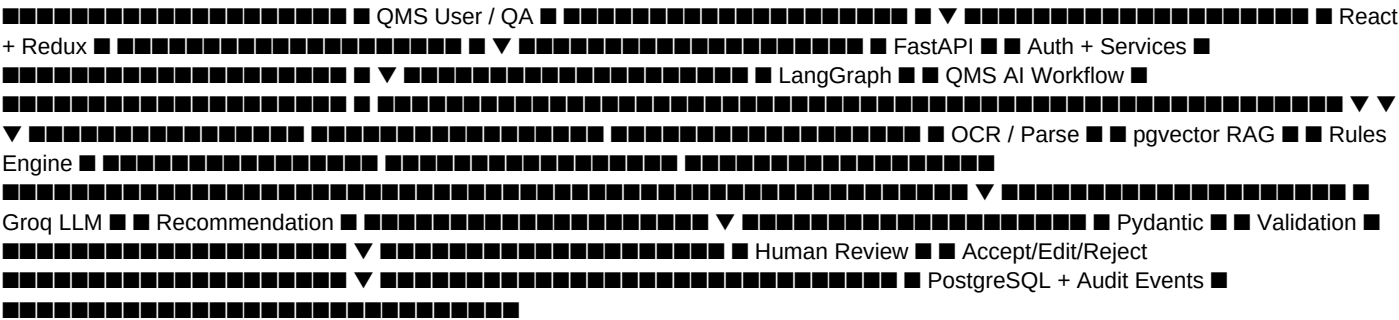
The challenge asks for a 5–10 minute QMS understanding video. The AI workflow should support one coherent story rather than a tour of every screen.

00:00–00:45 Open Dashboard. Show API batch API-2026-041 and current quality state. 00:45–01:30 Open in-process checks. Show temperature 84°C vs approved 76–80°C. 01:30–02:30 Create deviation. Upload/source document. Show OCR/extraction. Show AI recommendation + evidence. Human accepts/edits. 02:30–04:00 Open Investigation. Show AI suggested evidence/tasks. Review evidence. Move to Root Cause. 04:00–05:15 Show 5 Whys AI draft. Edit/confirm final root cause. 05:15–06:30 Create CAPA. Show AI corrective/preventive suggestions. Select and assign actions. 06:30–07:30 Show effectiveness. Monitor five batches. Show AI summary. Human marks effective. 07:30–08:15 Close CAPA/deviation. Show audit trail. 08:15–09:00 Open Batch Release. Show how the quality workflow informs batch disposition. 09:00–10:00 Show complaint/supplier links and summarize the connected QMS.

This sequence demonstrates understanding of the process while showing AI at the points where it actually adds value. It also avoids spending half the presentation clicking through settings pages, a beloved human tradition.

25. Implementation Priority

26. Final AI Architecture in One Diagram



27. Final AI Design Rule

The AI is the quality assistant, not the quality authority. Its job is to turn messy documents and connected QMS records into structured facts, relevant evidence, useful hypotheses and actionable drafts. The system of record remains the reviewed database state, and every important AI-assisted decision leaves an auditable trail.

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AI principle	Implementation decision
AI assists quality work	AI extracts, retrieves evidence, summarizes and recommends. It does not silently make final quality decisions.
Evidence first	Recommendations should expose the source documents/chunks used whenever retrieval is involved.
Structured outputs	Pydantic models validate AI JSON before it reaches business logic.
Human review	Severity, impact, root cause, CAPA actions and closure remain reviewable human decisions.
Auditability	Store model/prompt metadata, evidence, recommendation, reviewer decision and edits.
Reuse Round 1	Preserve existing OCR, structured extraction, RAG, risk assessment, LangGraph and Grog direction.

Stage	AI output	Human checkpoint
Deviation intake	Structured event extraction	Confirm extracted facts
Severity / impact	Recommendation + rationale	Accept/edit/reject
Investigation planning	Suggested evidence/tasks	Choose scope/tasks
Evidence review	Relevant source summaries	Review evidence
Root cause	Candidate causes / 5 Whys draft	Confirm final root cause
CAPA	Draft corrective/preventive actions	Approve action/owner/due date
Effectiveness	Trend/monitoring summary	Decide effective/ineffective
Closure	Draft closure summary	Authorize closure

Extracted field	Example	Confidence
title	Reactor temperature exceeded limit	0.98
batch_number	API-2026-041	0.99
process_step	Step 3 — Reaction	0.94
parameter	Temperature	0.98

Extracted field	Example	Confidence
expected_range	76–80 °C	0.96
actual_value	84 °C	0.99
duration_minutes	18	0.91
immediate_action	Cooling initiated; batch held	0.88

Node	Input	Output	Failure behavior
validate_input	file/context	validated request	Return validation error
parse_document	file ref	raw_text	Retry/parser fallback
ocr_fallback	image/scanned PDF	raw_text	Controlled OCR error
extract_fields	raw_text + context	structured extraction	Pydantic validation
validate_extraction	extraction	validated extraction	Ask repair once or fail
build_query	extraction	search query	Use safe fallback query
retrieve_evidence	query + metadata	evidence list	Return evidence-unavailable state
deterministic_assessment	extraction	rule results	Continue without LLM if rules fail
llm_assessment	rules + evidence	recommendation	Retry bounded; human review if unavailable
investigation_assist	investigation + evidence	task suggestions	No task created automatically
root_cause_assist	evidence + problem	RCA draft	Human review required
capa_assist	root cause + factors	action suggestions	Human review required
effectiveness_summary	monitoring data	summary	Human decision required
closure_summary	full record	draft summary	Human approval required
audit_ai_decision	recommendation + decision	audit event	Audit failure should be visible and retried

Prompt	Purpose	Example version
deviation-extraction	Structured field extraction	deviation-extraction-v2
deviation-triage	Severity/impact recommendation	deviation-triage-v2
investigation-planner	Evidence/task suggestions	investigation-planner-v1
root-cause-draft	5 Whys/candidate cause draft	root-cause-v1
capa-draft	Corrective/preventive suggestions	capa-v1
effectiveness-summary	Monitoring summary	effectiveness-v1
closure-summary	Closure draft	closure-v1

Risk	Guardrail
Hallucinated facts	Prompt says use only supplied evidence; validation + source IDs
Wrong severity	Human Accept/Edit/Reject before final persistence
Wrong root cause	AI output explicitly labeled candidate/draft
Invented CAPA owner/due date	Do not generate unless supplied context supports it
Unapproved document	Filter RAG retrieval to appropriate document status/version
Model outage	Return controlled degraded result; never fabricate completion
Prompt injection in documents	Treat document text as untrusted data, not instructions
Sensitive data leakage	Do not send unnecessary user/company secrets to model
Audit gap	Persist AI interaction and reviewer decision

Stored item	Example
request_id	req-uuid
entity	deviation / investigation / capa
stage	triage / root_cause / capa
provider	Groq
model	configured model name
prompt_version	deviation-triage-v2
input hash	sha256...

Stored item	Example
retrieved evidence IDs	SOP chunk IDs
recommendation	JSON
confidence	0.87
human decision	accepted/edited/rejected
edited fields	severity
created_at	timestamp

Test	Expected result
Clean deviation PDF	High-quality extraction + evidence + recommendation
Scanned deviation PDF	OCR fallback works
Missing batch number	Field remains missing; user can select batch manually
Out-of-range numeric value	Deterministic rule flags it
Weak RAG match	AI does not claim unsupported SOP evidence
Prompt injection text in PDF	Ignored as document instruction
LLM timeout	Controlled error/degraded workflow
User edits severity	Final record uses human value and logs edit
Root cause draft	Shows candidate/hypothesis language
CAPA suggestion	No automatic owner/due date unless supported
Closed record	AI cannot silently reopen or mutate it

Priority	AI capability	Build depth
P0	Deviation extraction + OCR fallback	Real
P0	RAG evidence retrieval	Real
P0	Severity/impact recommendation	Real + human review
P0	Investigation evidence/task assistant	Real but lightweight
P0	Root-cause / 5 Whys draft	Real + human review
P0	CAPA action suggestions	Real + human review
P1	Effectiveness summary	Light
P1	Closure summary	Light
P1	AI interaction/audit persistence	Real
P2	Advanced similarity to historical deviations	Only if time remains